

Reading a spread ladder

Kalshi does not quote a spread. It quotes a whole probability distribution, one strike at a time — the tool uses that spread ladder to determine if the Kalshi market implies an edge compared to the line from Vegas. Here is an example for the Georgia vs Arkansas game on September 19th, 2026.

Georgia is favored by 24.5 pts.

A sportsbook gives you one number: Georgia -24.5 . Kalshi instead runs a separate market on “Georgia wins by over 1.5”, “over 2.5”, “over 3.5” and so on up the board. For the purposes of this article, we will refer to each of these markets as a rung (rung on the spread ladder). Each price is a probability. Stack them and you have the whole distribution of the game's margin. Let x represent the points the home teams wins by.

the one idea

$P(x)$ = the chance the final margin comes in above x

A spread is a single point on that curve — the x where $P(x) = 0.5$. The ladder gives every point, so the chance of covering *any* number is read off the market rather than assumed.

The whole example below is **Georgia at Arkansas**, one snapshot taken from Kalshi's market a few days before the actual game on September 16th, 2026. Arkansas is the home team. The bet is **Georgia -24.5** — equivalently Arkansas $+24.5$ — ordinary betting notation, minus next to whoever gives the points. This means Georgia is heavily favored and Vegas expects them to win by 25 points.

One coordinate convention, and it is not betting notation. Internally every number is margin = home score – away score. With Arkansas home, -24.5 means Arkansas *receives* 24.5. It has to be a single fixed axis or $P(x)$ could not be shifted, compared or integrated — “home” just picks which direction is positive, the way a graph picks which way is up.

The order book, and what the tool ignores

Open *Georgia over 41.5* on Kalshi and the book is a mess. Here it is, the same snapshot, everything with at least one contract resting:

YES BIDS – BUY “OVER 41.5”

PRICE	CONTRACTS	\$
13¢	120	16
12¢	400	48
8¢	790	63
3¢	1,046	31
2¢	1,961	39
1¢	2,800	28

NO BIDS – BUY “UNDER 41.5”

PRICE	CONTRACTS	\$
83¢	64	53
59¢	4,852	2,863
44¢	11,483	5,053
30¢	42,670	12,801
15¢	149,152	22,373
1¢	3,430,820	34,308

Why there is no separate ask column

Kalshi never sells you a contract; it matches you with somebody taking the other side. So a YES *ask* is not a third list — it is a NO bid, quoted from the other end. Exactly one contract settles at \$1, so the two prices are tied together:

one contract pays \$1, and exactly one side wins

YES price + NO price = \$1

so every NO bid in the right-hand table is a YES ask

NO bid 83¢ = 100¢ – 83¢ = **YES ask 17¢**

NO bid 59¢ = YES ask 41¢, and so on down the book

Fold the two tables together and the strike is one ordinary two-sided market. Both columns below are the same book, read from opposite ends:

STRIKE 41.5	YES – “OVER 41.5”	NO – “UNDER 41.5”
best bid	13¢	83¢

STRIKE 41.5	YES – “OVER 41.5”	NO – “UNDER 41.5”
best ask	17¢	87¢
bid + bid	96¢ – 4¢ short of a dollar	
ask + ask	104¢ – 4¢ over	

That 4¢ gap is the whole cost of dealing here. Buy YES at 17¢ and NO at 87¢ and you have spent 104¢ to be certain of getting \$1 back — a guaranteed 4¢ loss, which is the market maker's edge. Sell both sides into the bids and you collect 96¢ for a \$1 liability: the same 4¢, pointing the other way.

Note the 2,800 contracts resting at 1¢ on the YES side, and the far larger block opposite it: **3.4 million contracts bid at 1¢**, \$34,308 of resting orders — which, read the other way, is 3.4 million contracts offering to sell YES at 99¢.

Neither is a valuation. A YES bid at 1¢ pays \$1 if Georgia wins by 42 or more — something this market prices at 15¢. Nobody bids one cent for a fifteen-cent thing expecting fair value. It is a resting option, sitting there in case somebody market-sells into a thin book during a chaotic moment in the fourth quarter. It is a bet on *someone else's panic*, not a belief about the football game.

This tool never sees any of it. It reads two fields per strike, the best bid and best ask, top of book and nothing else. On this strike that is 0.13 / 0.17. The 1¢ orders sit five levels down and are invisible, as is the 3.4 million on the other side. The tool ignores the junk bids and asks.

What happens if the junk *becomes* the top of book

That is the case worth defending against. If every real bid gets pulled, the 1¢ order is the best bid and the market quotes 0.01 / 0.17 — a 16¢ spread. Five filters stand in the way for each spread market on a game, and they run in this order:

1. hard width cap -- `MAX_STRIKE_WIDTH = 0.15`
`width 0.16 > 0.15 => strike dropped entirely`
2. never traded, and wider than typical for this game
`open_interest <= 0 AND width > median_width => dropped`
3. a quote spanning the whole range says nothing
`bid <= 0 AND ask >= 1 => dropped`
4. survivors are weighted, not treated alike
$$w = (1 / \text{width}) \times (1 + \ln(1 + \text{open_interest}))$$
5. and the leftover disagreement is measured, not assumed away
the band -- derive the whole curve from bids, then from asks

Open interest is traded contracts, not resting ones. It counts positions that have actually been opened — matched, filled, held. The 1¢ orders above are offers to make a contract, not contracts, so none of them reach the weight: this strike shows 6,820 open interest against 3.7 million contracts resting in the book.

Bad Spread Ladders: Ideally a spread ladder should always have better odds for a more likely scenario. Georgia winning by 41.5 pts is more likely rung than Georgia winning by 45.5. Sometimes in thin markets, however, these extreme rungs share the same odds, or have wild bid-ask spreads. These filters account for mispriced rungs and clean up the ladder.

Step 2 below walks the rungs and, wherever a run of them goes the wrong way, replaces that whole run with *one* value — their average, weighted by *w*. This can happen in two scenarios:

crossed: a further rung priced higher than a nearer one

-14.5 P = 0.2450 w = 201 3c wide, 153 OI

-13.5 P = 0.2600 w = 327 2c wide, 254 OI — impossible, P rose

=> both become $(0.2450 \times 201 + 0.2600 \times 327) / 528 = \mathbf{0.2543}$

duplicate: two rungs quoting the same odds

-6.5 and -5.5 both at P = 0.1100 => already flat, nothing to pool

So the weight decides *where between them* the pooled value lands. A rung quoted 1¢ wide with 50,000 contracts is weighted far greater than one quoted 4¢ wide and never traded. Pool those two and the answer sits essentially on the liquid one — the thin quote is pulled into line rather than dragging the curve with it. That is the whole job: keep the information in a bad rung without letting it outvote a good one.

On this snapshot of the Georgia vs Arkansas game, all 24 rungs survive the filters — the widest bid/ask spread is 4¢.

THE DATA

The whole ladder

Below is the snapshot for all 24 rungs, plus every derived column. The first six columns are what Kalshi returned; the rest you can rebuild from them.

THRESH	BID	ASK	MID	BID-ASK	OPEN INT.	X	P	PAVA WT	Z	FIT WT
41.5	0.13	0.17	0.1500	0.04	6,820	-41.5	0.8500	245.7	-1.7346	0.5100
38.5	0.16	0.18	0.1700	0.02	4,416	-38.5	0.8300	469.7	-1.5856	0.5644
35.5	0.22	0.23	0.2250	0.01	6,439	-35.5	0.7750	977.0	-1.2368	0.6975
33.5	0.28	0.29	0.2850	0.01	1,612	-33.5	0.7150	838.6	-0.9198	0.8151
30.5	0.35	0.36	0.3550	0.01	5,646	-30.5	0.6450	963.9	-0.5971	0.9159
27.5	0.45	0.46	0.4550	0.01	19,338	-27.5	0.5450	1087.0	-0.1805	0.9919
24.5	0.52	0.53	0.5250	0.01	49,660	-24.5	0.4750	1181.3	0.1001	0.9975
23.5	0.55	0.56	0.5550	0.01	8,328	-23.5	0.4450	1002.8	0.2209	0.9879
21.5	0.59	0.60	0.5950	0.01	9,061	-21.5	0.4050	1011.2	0.3847	0.9639
20.5	0.63	0.64	0.6350	0.01	2,701	-20.5	0.3650	890.2	0.5537	0.9271

THRESH	BID	ASK	MID	BID-ASK	OPEN INT.	X	P	PAVA WT	Z	FIT WT
17.5	0.68	0.71	0.6950	0.03	372	-17.5	0.3050	230.7	0.8236	0.8479
16.5	0.71	0.73	0.7200	0.02	379	-16.5	0.2800	347.0	0.9445	0.8064
14.5	0.74	0.77	0.7550	0.03	153	-14.5	0.2450	201.2	1.1255	0.7399
13.5	0.77	0.79	0.7800	0.02	254	-13.5	0.2200	327.1	1.2657	0.6864
11.5	0.80	0.82	0.8100	0.02	0	-11.5	0.1900	57.9	1.4500	0.6156
10.5	0.82	0.84	0.8300	0.02	164	-10.5	0.1700	305.2	1.5856	0.5644
9.5	0.82	0.86	0.8400	0.04	0	-9.5	0.1600	30.0	1.6582	0.5376
7.5	0.85	0.89	0.8700	0.04	23	-7.5	0.1300	104.2	1.9010	0.4524
6.5	0.88	0.90	0.8900	0.02	484	-6.5	0.1100	359.2	2.0907	0.3916
5.5	0.88	0.90	0.8900	0.02	39	-5.5	0.1100	234.6	2.0907	0.3916
4.5	0.89	0.90	0.8950	0.01	11	-4.5	0.1050	349.6	2.1429	0.3759
3.5	0.91	0.92	0.9150	0.01	11	-3.5	0.0850	349.7	2.3763	0.3111
2.5	0.92	0.93	0.9250	0.01	727	-2.5	0.0750	759.1	2.5123	0.2775
1.5	0.92	0.93	0.9250	0.01	14	-1.5	0.0750	370.5	2.5123	0.2775

Every rung is quoted on Georgia, the away team — nobody buys “Arkansas by over 10”, so Kalshi doesn't even offer it. A one-sided ladder is normal for a big spread, more evenly matched teams will have ladders on both sides of 0. Column by column, as Excel formulas:

```

given: threshold, bid, ask, open interest
mid      = (bid + ask) / 2
bid-ask  = ask - bid
x        = -threshold  away-team strike; a home strike would be +threshold
P        = 1 - mid      away-team strike; a home strike would be mid
PAVA wt  = (1/bid-ask) * (1 + LN(1 + OI))
z        = LN((1 - P) / P)    blank unless 0.05 < P < 0.95
fit wt   = 4 * P * (1 - P)    same window

```

From that table to a bet

01 Turn each price into a point on one curve

Following the notation we set, a price on Georgia has to be flipped before it can join a function measured from Arkansas's side (the home team).

```
chance Georgia wins by > 41.5 = 0.1500
=> chance Arkansas margin > -41.5 = 1 - 0.1500
x = -41.5, P = 0.8500

chance Georgia wins by > 2.5 = 0.9250
x = -2.5, P = 0.0750
```

Read it as: Arkansas losing by fewer than 41.5 is an 85% shot; losing by fewer than 2.5 is only 7.5%.

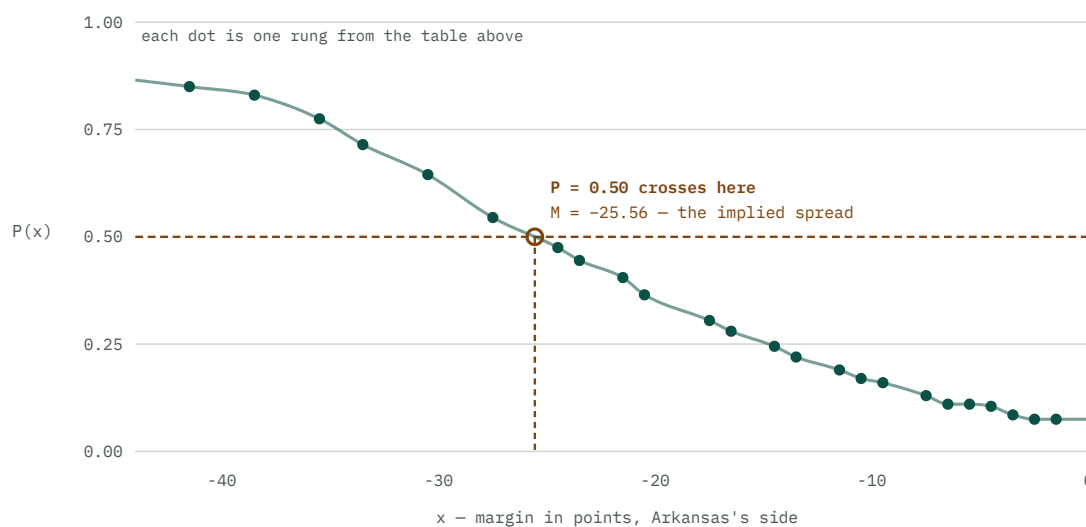
02 Force the sequence to be non-increasing

$P(x)$ can never rise as x rises — a bigger margin is always less likely. Thin quotes break that constantly, and ladders with bad rungs are filter out and replaced by their weighted average as describe above.

On this particular ladder it changes nothing. All 24 points already descend, so no ladder cleaning is needed.

The median is the implied spread

Here is the whole ladder as a picture — every rung from the table, plotted as x against P :



The 24 rungs and the monotone curve through them. It only has to cross $P = 0.50$ once, and where it does is the implied spread.

Find where it crosses one half. The two rungs straddling it:

$$P(-27.5) = 0.5450$$

$$P(-24.5) = 0.4750$$

straight-line interpolation

$$-27.5 + 3 \times (0.5450 - 0.5) / (0.5450 - 0.4750) = -25.5714$$

what the tool gets, using a monotone cubic

implied margin = -25.5600

A spreadsheet will land on -25.5714, not -25.5600. If you follow along in Excel you will get a slightly different median. The tool fits a Fritsch–Carlson monotone cubic through all the rungs rather than joining them with straight lines — chosen because it cannot overshoot. Everything downstream from here is reproducible in Excel; this one step is not.

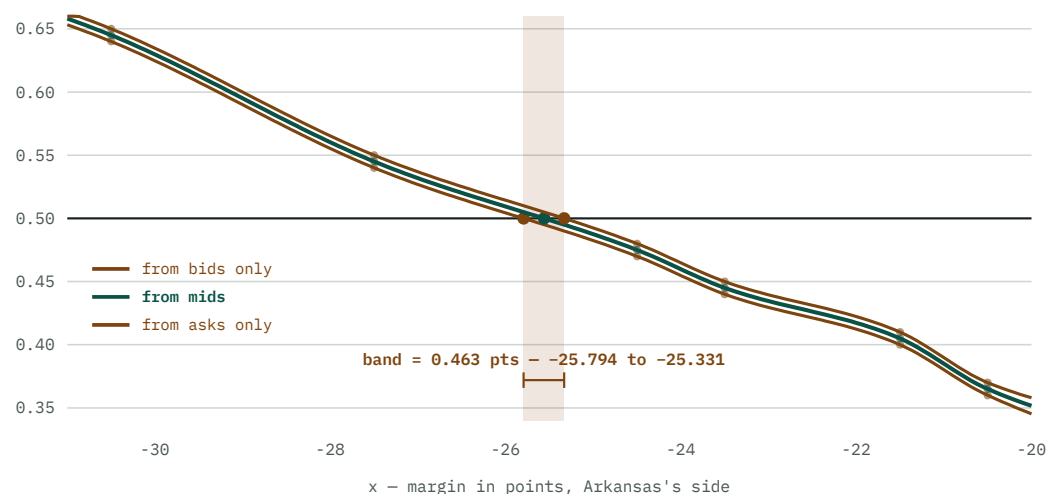
Do it twice more to measure the market's own uncertainty

Rebuild the entire curve from bid prices only, then from ask prices only. On an away-team strike the mapping inverts, so the *ask* column produces the lower margin:

$P = 1 - \text{ask on every strike} \Rightarrow \text{median} = -25.794$

$P = 1 - \text{bid on every strike} \Rightarrow \text{median} = -25.331$

band = 0.463 pts



The same rungs, priced three ways, zoomed on the crossing. Each curve meets $P = 0.50$ at its own x ; the shaded strip between the outer two is the band.

The band is the honest version of “how sure is the market”. A plain bid-ask filter misses games whose *wide* strikes happen to sit right at the median; this is more robust, because it is measured at the crossing, which is where the number is actually read.

This tool will grade every Spread Ladder

The band leads the grade, but it does not decide it alone. A ladder can be quoted a cent wide and still be worthless if nobody has traded it, so each grade sets three bars at once and a market has to clear *all three*:

GRADE	BAND	OPEN INTEREST	RUNGS TRADED	MONEY
S – Deep market	≤ 0.5	≥ 10,000	≥ 60%	≥ \$500k

GRADE	BAND	OPEN INTEREST	RUNGS TRADED	MONEY
A – Tight market	≤ 0.5	≥ 10,000	≥ 60%	–
B – Readable market	≤ 1.0	≥ 2,000	≥ 30%	–
C – Loose market	≤ 2.0	≥ 500	any	–
D – Unreadable	> 2.0	< 500	any	–

The grade then limits the call. A Loose market cannot produce anything stronger than a lean however large the edge, because that edge was measured against a number the market itself is unsure of — and only S and A markets count toward the headline accuracy record. **The grade describes the market, not the bet.** A Deep market can still offer nothing worth betting, and frequently does.

Georgia at Arkansas, this snapshot

band 0.45 ≤ 0.5 ✓

open interest 151,847 ≥ 10,000 ✓

rungs traded 100% ≥ 60% ✓

dollar volume \$162,900 < \$500k ✗

grade A – Tight market short of S on money alone

Fit the width

Defining some important columns shown in the spread ladder table above:

x a margin in points. $x = -41.5$ means "Georgia by 41.5".
 P a probability. $P(x)$ = the chance the margin beats x .
 M the median, where $P = 0.5$. This IS the implied spread.
 W the width: how far margins scatter around M, in points.

So we know the median ($M=25.56$) and the Vegas line is 24.5, so our spread ladder is giving us an edge of 1.06. Now we want to determine what that edge means in terms of probability, and that is given to us via the *steepness* of the curve. Since $P(x)$ counts everything above x , its slope is the probability density — at any x , the slope is the chance the game finishes within a point of there.

Below we will determine the *width* which will later be used to determine the standard deviation of the data and its implied probability.

Curve Fitting using Logistics equation

As seen above, the spread ladder is plotted as a logistic. The formula has exactly two knobs:

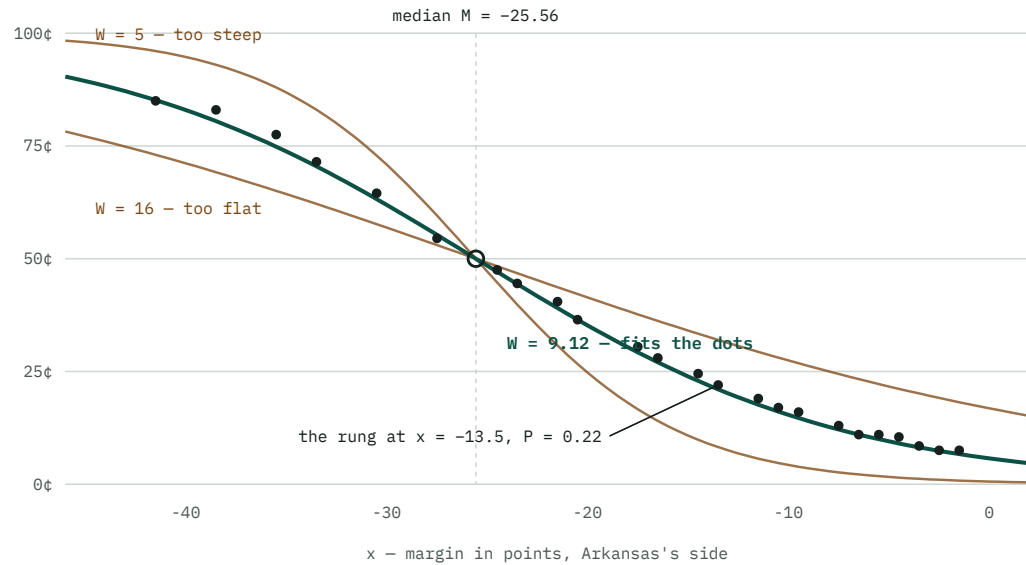
$$P(x) = 1 / (1 + e^{((x - M)/W)})$$

M shifts it left or right. W makes it steep or flat.

Step 3 already fixed $M = -25.56$, which leaves a whole *family* of candidate curves — one for every possible W . The job is finding which member the market is quoting.

The goal to find a W that best fits the data from Kalshi's spread ladder.

For example, the rung at $x = -13.5$ trades at 78¢, so $P = 0.22$. Plugging that into the equation above and $W = 9.53$.



Three candidates, all pinned at the median. Near it they lie on top of one another; out in the tails they fan wide apart. The 24 dots are this game's actual prices.

The far rungs impact the width

Rearranged for W, any single rung paired with the median already determines the width:

$$P = 1 / (1 + e^{((x - M)/W)}) \Rightarrow W = (x - M) / \ln((1 - P)/P)$$

X	P	X - M	Z	W IMPLIED
-41.5	0.8500	-15.94	-1.7346	9.189
-38.5	0.8300	-12.94	-1.5856	8.161
-35.5	0.7750	-9.94	-1.2368	8.037
-33.5	0.7150	-7.94	-0.9198	8.632
-30.5	0.6450	-4.94	-0.5971	8.273
-27.5	0.5450	-1.94	-0.1805	10.749
-24.5	0.4750	+1.06	+0.1001	10.591
-23.5	0.4450	+2.06	+0.2209	9.326
-21.5	0.4050	+4.06	+0.3847	10.554

X	P	X - M	Z	W IMPLIED
-20.5	0.3650	+5.06	+0.5537	9.138
-17.5	0.3050	+8.06	+0.8236	9.786
-16.5	0.2800	+9.06	+0.9445	9.593
-14.5	0.2450	+11.06	+1.1255	9.827
-13.5	0.2200	+12.06	+1.2657	9.529
-11.5	0.1900	+14.06	+1.4500	9.696
-10.5	0.1700	+15.06	+1.5856	9.498
-9.5	0.1600	+16.06	+1.6582	9.685
-7.5	0.1300	+18.06	+1.9010	9.500
-6.5	0.1100	+19.06	+2.0907	9.116
-5.5	0.1100	+20.06	+2.0907	9.595
-4.5	0.1050	+21.06	+2.1429	9.828
-3.5	0.0850	+22.06	+2.3763	9.283
-2.5	0.0750	+23.06	+2.5123	9.179
-1.5	0.0750	+24.06	+2.5123	9.577

Two things follow from that table, and they are the reason the ladder exists at all. The first is visible in the chart above: near the median the candidate curves are stacked on top of one another, so a rung there barely discriminates — any width fits it about equally well. Out at -41.5 they are separated by 20¢, so a price there picks one decisively.

Before the fit: where the 8.5 comes from

The fit below leans on a prior width of 8.5. That number is not from this game and not from Kalshi — it is a standing assumption about college football, carried in the same units as W . Converted to the scale it is easier to argue about:

$$\text{prior } W = 8.5 \Rightarrow SD = 8.5 \times 1.8138 = \mathbf{15.4 \text{ pts}}$$

Season on season, FBS margins scatter around the closing spread with a standard deviation of roughly **13 to 17 points**. 8.5 sits in the middle of that. It is deliberately unopinionated: near enough to be harmless when it is wrong, and far enough from the absurd to catch a ladder that has gone wrong.

So the prior gets a vote, sized in the same currency as the data: it counts for **five rungs** of information. Fitted widths across a full slate land between 7.6 and 10.9, so five rungs is enough to rescue a ladder with nothing to say and far too little to overrule one that does. This game supplies 15.6 effective rungs, so the prior holds about a quarter of the vote.

Behind it sits a hard clamp of W between 6 and 12 — SD 10.9 to 21.8 points. On a measured slate it never binds. If it starts binding, the fit is wrong, not the game.

The fit

So all twenty-four rungs go in, weighted. Using the `z` and `fit_wt` columns from the ladder:

```
weighted least squares through the origin
raw W =  $\Sigma(\text{wt} \cdot z \cdot (x - M)) / \Sigma(\text{wt} \cdot z^2)$ 
      = 234.1600 / 25.1209
      = 9.3213

how much signal the fit had
info =  $\Sigma \text{wt} = 15.6470$  ← effective rungs, of 24
k     =  $15.6470 / (15.6470 + 5) = 0.7578$ 

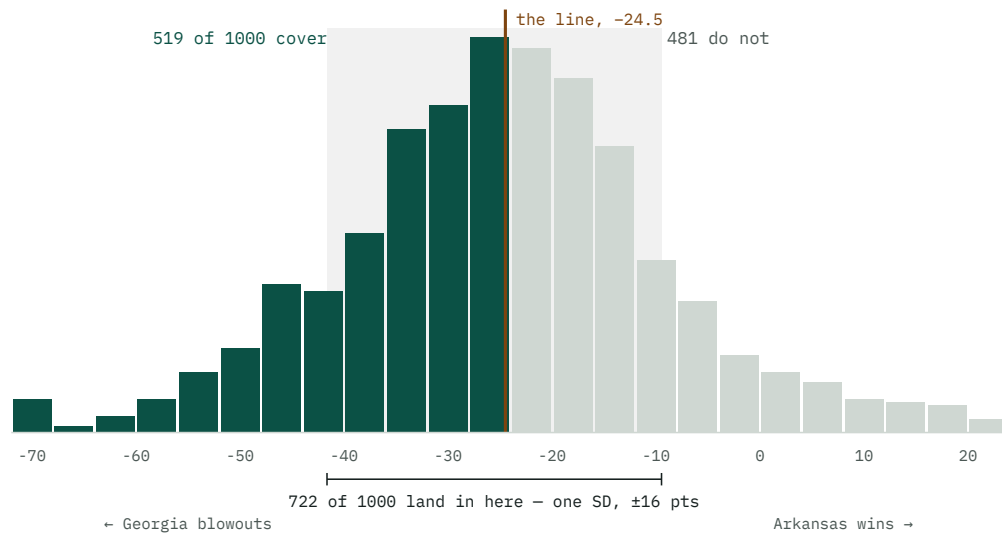
shrink toward the 8.5 prior
W =  $0.7578 \times 9.3213 + 0.2422 \times 8.5 = \mathbf{9.1224}$ 
```

```
in Excel, with x, z and fit_wt as columns
raw = SUMPRODUCT(fit_wt, z, (x - M)) / SUMPRODUCT(fit_wt, z, z)
info = SUM(fit_wt)
W     = (info/(info+5))*raw + (5/(info+5))*8.5
```

The fit weight $4P(1-P)$ is what makes it a weighted regression rather than a plain one: rungs near the median, whose z is least reliable, count least. The prior then pulls the raw 9.3213 down to 9.1224 — 0.2 points of insurance on a fit that did not need rescuing.

What $W = 9.12$ means

Simulate a thousand replays of this game from the market's own numbers — centre -25.56 , width 9.12 :



1,000 replays. Each bar is a four-point bucket of final margins. Shaded bars are results where Georgia covers -24.5 .

The standard deviation is derived from W by a fixed constant.

$$\text{standard deviation} = W \times \pi/\sqrt{3} = 9.1224 \times 1.8138 = \mathbf{16.55 \text{ pts}}$$

Read it off the picture rather than the formula: 722 of the 1,000 replays land within 16.55 points of the centre and the rest spill out either side.

Note that the shape is not a bell curve — it is squat and long-tailed. Eleven percent of replays are Georgia by 45 or worse and six percent are Arkansas outright, which is precisely what those far rungs were pricing.

The second reading of W is the one the rest of the tool runs on. **Per point** is the conversion rate: shift the line one point, and this is how much the chance of covering moves. Differentiate the logistic at its centre and it is exactly $1/(4W)$ — but it is easier to just count it off the replays:

Georgia covers -24.5 **519 of 1000**

Georgia covers -25.5 **488 of 1000**

31 games switch sides over that one point

$1/(4W) = 2.74\%$ predicts 27 per 1000; the gap is sampling noise

So at this width one point of spread is worth 2.74 percentage points of win probability. Wider game, flatter curve, fewer games moved per point. That number is the exchange rate every later step trades through. Now that we have extracted data from Kalshi's spread ladder let's look at the money line market to support it.

The moneyline is just the rung at $x = 0$

Kalshi runs a separate market on “Arkansas wins?”. Arkansas winning is Arkansas's margin being greater than zero. So the money line is one more point on the exact same curve. In this case the rung the spread ladder does not have a rung sitting at $x = 0$ since it is such a lopsided game. As of the snapshot, Kalshi is trading Arkansas wins at 4.5 cents.

```
the ladder's rungs run -41.5 ... -1.5
the moneyline supplies the one at  $x = 0$ 
```

```
fitted curve says  $P(0) = 0.0572$ 
moneyline market says      = 0.0450
```

Two markets on the same exchange, quoting the same event, disagreeing by 1.2 cents. The question is what that is worth *in points*, because points is the only scale on which it can be argued with the ladder.

So ask it backwards: where would the median have to sit for $P(0)$ to equal 4.5%? Set $x = 0$ in the logistic and solve for the centre:

$$P(0) = 1 / (1 + e^{((0 - M)/W)}) \Rightarrow M = -W \cdot \ln((1 - P(0)) / P(0))$$

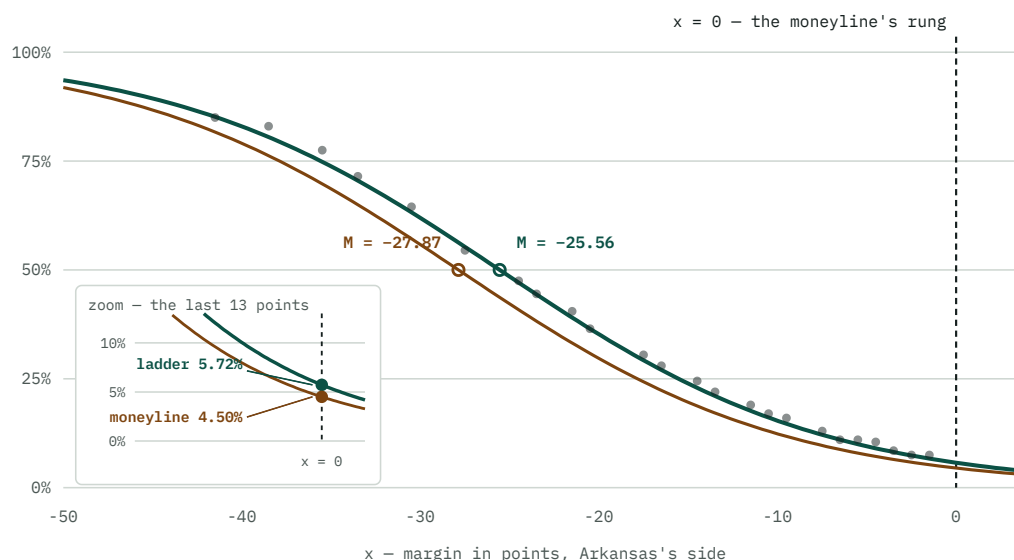
$$\text{mid } 0.045 \quad -9.1224 \cdot \ln(0.955/0.045) = \textbf{-27.8695}$$

$$\text{bid } 0.040 \quad = -28.9916$$

$$\text{ask } 0.050 \quad = -26.8605$$

$$\textbf{moneyline band = 2.1311 pts}$$

The ladder says Georgia by 25.56; the moneyline is priced as if the number were Georgia by 27.87. Which is the picture below: one curve, slid left until it passes through the moneyline's price at the rung the ladder does not have.



The grey dots are the 24 spread rungs; the solid curve is the fit through them. At $x = 0$ it reads 5.72%. The moneyline pays at 4.50%, and the only way to make the same curve say 4.50% there is to move its median from -25.56 to -27.87.

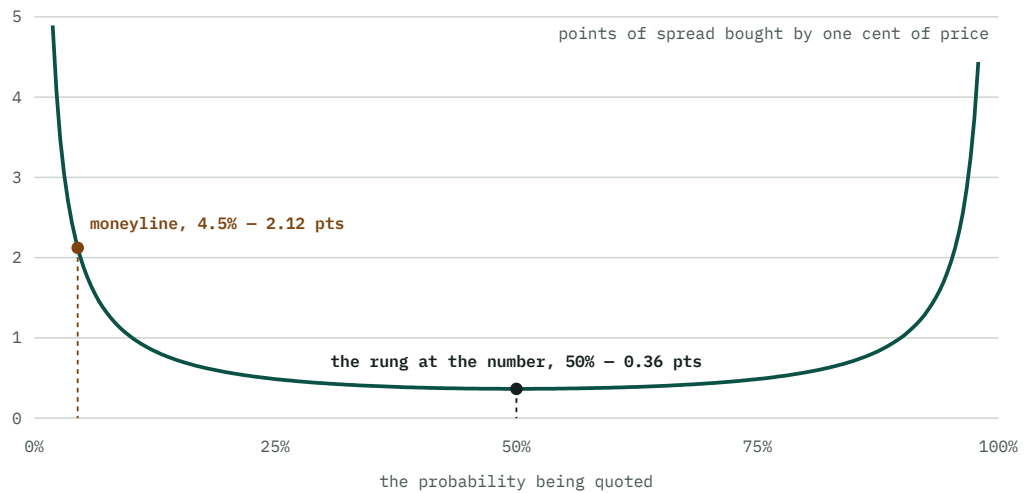
Note what the conversion needed. A win probability alone cannot give you a spread — 4.5% is equally consistent with a 27-point favorite in a normal game and a 15-point favorite in a freakishly predictable one. You have to know how far margins scatter, which is W (width) from step 5. That is why it is worth fitting properly: it is not only the cover-chance number, it is the exchange rate the we use to determine the implied spread from the money line.

The conversion is least trustworthy on the most lopsided games. Deep in the tail the mapping from probability to points is steep, so a single tick of quoting noise moves the answer a long way. Take Arkansas from 4.5¢ to 5.5¢ — one cent, the smallest move Kalshi allows — and the implied number travels from Georgia by 27.87 to Georgia by 25.94, nearly two points. On a near-even game the same cent is worth about a third of a point, because at 50% the curve is at its flattest. The moneyline is therefore an imprecise witness on exactly the games where it sits furthest from the spread. The next step prices that imprecision rather than arguing with it.

$$\text{points per cent} = W \times 0.01 / (P \times (1 - P))$$

at the number, $P = 0.50$ $9.1224 \times 0.01 / 0.2500 = \mathbf{0.36 \text{ pts}}$

at the moneyline, $P = 0.045$ $9.1224 \times 0.01 / 0.0430 = \mathbf{2.12 \text{ pts}}$



The same one-cent tick, valued across the curve. Near the number it barely moves the spread; out where a heavy favourite's moneyline trades it moves it six times as far.

So the moneyline is imprecise on this game *because* it is a blowout, not because the market is thin — it carries 48,000 contracts of open interest. A cent of quoting noise at 4.5% simply is worth two points of spread. The next step prices that imprecision rather than arguing with it.

Combine the spreads by precision

Two markets now have an opinion about the same number, and they disagree: the ladder says Georgia by 25.56, the moneyline says Georgia by 27.87. Averaging them would be wrong, because one of those readings is far better measured than the other. We want to find a composite spread. Two symbols do the work:

σ sigma — how precisely a market pins the fair spread, in points
 w weight — $1/\sigma^2$, that market's share of the vote

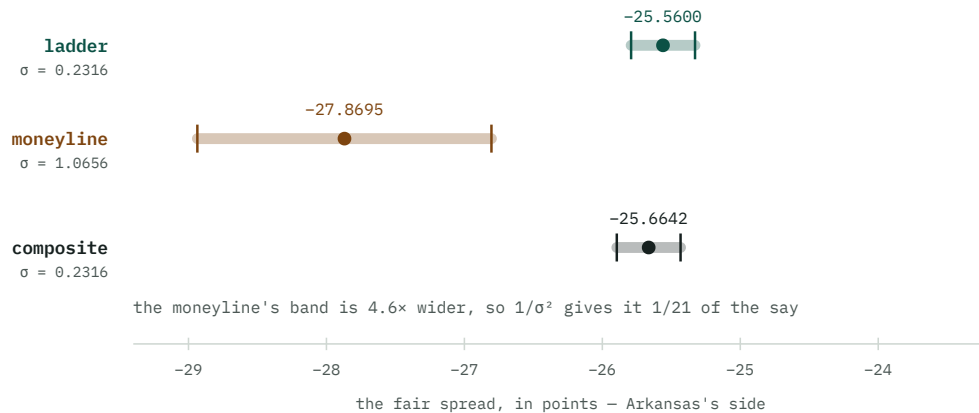
Three symbols here, and two of them look alike. Step 5's w is the width of the *game*: 9.12 points, how far margins scatter. The *prec* below is the weight of a *quote*, and it measures something else entirely — how much to trust one market against the other. And σ is a third thing again: how far the quotes leave the number itself in doubt. A game can land 16 points off a spread that is nonetheless known to a quarter of a point, so a wide game and a sharp price are not in tension.

Each market's σ is derived from half of the band.

step 4 — the median from bids only vs from asks only
 ladder band 0.4632 pts $\sigma = 0.4632/2 = \mathbf{0.2316}$

step 6 — the 4¢ bid and the 5¢ ask, converted to points
 moneyline band 2.1311 pts $\sigma = 2.1311/2 = \mathbf{1.0656}$

Halving is what turns an interval into a distance: the band runs from one edge to the other, while σ is the distance from the middle out to one edge. Read the two markets as error bars on the same axis:



Both markets are estimating one quantity. The ladder's bar is narrow, the moneyline's is 4.6 times wider, and the composite lands almost on top of the narrow one.

Now the weighting. Precision is $1/\sigma^2$, and the square is not decoration: it is what makes a market 4.6 times sharper count **21 times** as much rather than 4.6 times. That is the weighting that minimises the error of the blend when two noisy readings measure the same quantity.

```

each market's precision, from its own  $\sigma$ 
ladder       $\sigma = 0.2316$   prec =  $1/0.2316^2 = 18.6427$ 
moneyline  $\sigma = 1.0656$   prec =  $1/1.0656^2 = 0.8807$ 
                                total = 19.5234

the composite is their average, weighted by prec
(18.6427  $\times$  -25.5600) + (0.8807  $\times$  -27.8695) = -501.0274
-501.0274 / 19.5234 = -25.6642

```

In this case, the moneyline's implied spread of -27.86 is way less impactful on the composite than the spread ladders -25.5 spread. On a close game, the moneyline's implied spread will have more of an impact on the composite spread.

One number remains, the composite's own σ , which step 8 needs. The tool keeps the best input's σ and claims no improvement it cannot justify:

```

 $\sigma = \min(0.2316, 1.0656) = 0.2316$ 

```

Weigh in Vegas's line

Why shrink toward the very number you are betting against? Because the alternative is a worse assumption. Giving Vegas no weight is a standing claim that on every game a Kalshi ladder knows something a much larger, older and more specialized market has missed. Letting it in makes the tool demand evidence in proportion to how sharp Kalshi actually is: a razor-tight ladder overrides Vegas almost entirely, a loose one barely moves off it.

Vegas enters with an *assumed* σ of 1.5 points, the one number in this whole chain that is not measured from quotes. Closing CFB spreads sit within roughly a point of true consensus, so 1.5 is deliberately loose: informative enough to catch a lone bad Kalshi read, weak enough that a tight ladder dominates it.

```
both sides weighted on the same definition of  $\sigma$ 
composite  $\sigma = 0.2316$  from step 7
Vegas  $\sigma = 1.5000$  assumed
```

```
k = (1/0.23162) / (1/0.23162 + 1/1.52)
    = 18.6433 / (18.6433 + 0.4444) = 0.9767
```

```
estimate = 0.9767 × -25.6642 + 0.0233 × -24.5
estimate = -25.6371
```

Step 3 gave -25.56: the spread ladders implied spread. The moneyline pushed it out to -25.66, and Vegas pulled 2.3% of the way back, to -25.64. Both moves are small, and for the same reason: every opinion arrives weighted by its own precision, and on a Tight market the ladder is much sharper.

Then three gates, in order, any one of which ends it.

```
1. is the line inside the market's own band?
-24.5 vs [-25.794, -25.331] - outside by 0.83, continue
```

That interval is not built around the estimate. It is step 4's two crossings: the median implied by bids only, and the median implied by asks only. Had Vegas's line sat in between the bid/ask spread of Kalshi's market, the apparent disagreement would have been smaller than the cost of dealing —

nothing to bet even if the direction were right. -24.5 sits 0.83 points outside, so the disagreement is real rather than bookkeeping.

```
2. is the edge bigger than the vig?  
edge = -25.6371 - (-24.5) = -1.1371  
1.1371 ≥ 1.0 - continue
```

Gate 1 only established that the two markets disagree. This one asks whether the disagreement is large enough to pay for the bet.

```
3. how big, and is the market good enough to say so?  
1.1371 < 2.0 => lean
```

The last gate sizes the call — under 2 points a lean, 2 to 3.5 a solid, above that a strong — and caps it. A Loose market cannot produce anything above a lean however large the edge looks, because that edge was measured against a number the market itself is unsure of.

```
and the cover probability, off the fitted distribution  
P(Arkansas covers -24.5) = 1 / (1 + e^((-24.5 - -25.6371)/9.1224))  
= 0.4689  
P(Georgia covers -24.5) = 53.1%
```

THE PICK

Georgia -24.5 — lean. Two Kalshi markets put the fair number at Georgia by 25.7; Vegas offers 24.5. A 1.14-point edge, worth 53.1% to cover against the 52.4% a -110 line demands. Thin — which is exactly why it is a lean and not a strong.

WHAT IT IS WORTH

Pricing the cover probability - in depth.

Step 8 calculates Georgia by 25.64: an edge of **1.14 points**. Step 5 already fixed the exchange rate — 2.74 percentage points of win probability per point of spread — so converting that edge is one line:

chance of covering $\approx 50\% + (\text{edge in points}) \times (\text{per point})$

$= 50\% + 1.14 \times 2.74\%$

$= \mathbf{53.11\%}$ the exact logistic also gives 53.11%

So Georgia covers -24.5 about 53.11% of the time according to this tool. The only question left is what the bet costs. A -110 price means risking \$110 to win \$100, so it breaks even at:

$p \times 100 = (1 - p) \times 110 \Rightarrow 210p = 110$

$p = \mathbf{110 / 210 = 52.38\%}$

53.11% against 52.38%. The bet is on, by three quarters of a percentage point. 53.11% is fair value at -113, so -110 is worth taking and anything shorter than -113 is not:

fair price for a 53.11% chance

american odds $= -(0.5311 / 0.4689) \times 100 = \mathbf{-113}$

-110 offered needs 52.38% have 53.11% - **bet**

-115 offered needs 53.49% have 53.11% - **no bet**

Three quarters of a point of probability is a thin margin, and it is thin because the edge is thin — not because anything upstream is shaky. That is precisely what a *lean* means, and why two points of the same edge would be a solid.

NOTE: Kalshi's market fees are not incorporated into this tool.

How much does one far rung actually matter?

The rung at $x = -41.5$ is the market saying Georgia wins by 42-plus about 15% of the time. Reprice it by a full ten cents, a large move on a market quoting 15¢, and refit across all 24 rungs:

-41.5 as quoted, 15c -> W = 9.12 -> 2.74%/pt -> cover 53.11% **BET**
-41.5 at 20c -> W = 9.21 -> 2.71%/pt -> cover 53.08% **BET**
-41.5 at 25c -> W = 9.28 -> 2.69%/pt -> cover 53.06% **BET**

Ten cents on one rung moves the width by 0.16 and the cover chance by five hundredths of a percentage point. No single quote can hijack the answer, which is precisely what fitting across every rung and shrinking toward a prior bought. Read the width off two adjacent rungs instead and those same ten cents would *be* the answer.

What does decide it is the far rungs moving **together**: the market repricing how wild the game could get, rather than one strike drifting.

every rung 10+ pts out, 3c wider -> W = 10.06 -> 2.49%/pt -> cover 52.82% **BET**
every rung 10+ pts out, 8c wider -> W = 11.82 -> 2.12%/pt -> cover 52.40% **BET, barely**
every rung 10+ pts out, 12c wider -> W = 12.00 -> 2.08%/pt -> cover 52.37% **no bet**

The edge never moved in any of those rows — 1.14 points throughout. Only the bar did. And on the last row W stopped at 12.00 because that is the tool's built in clamp, a sign the ladder had stopped describing a football game.

THE OTHER 23 MARKETS

Is another rung a better bet?

Everything so far judged a single number. But the ladder is not one market — it is 24 separate outright markets, each with its own bid, ask and question. “Georgia by more than 35.5” is a real tradeable contract at 23¢. The same two ingredients price every one of them.

The recipe, on one rung

Take the rung at 35.5. Three lines:

1. the edge: how far our number clears this rung, in points
edge = $25.6371 - 35.5 = -9.86$ pts
negative: our number falls 9.86 short of 35.5

2. edge into probability, at 2.74%/pt
chance = $50\% + (-9.86 \times 2.74\%) = 23.0\%$
exact logistic: $1/(1 + e^{(9.86/9.1224)}) = 25.33\%$

3. what the market charges for the same claim
ask = 23¢ => break-even 23.0%, quoted as **+335**

25.33% against a 23.0% break-even
edge = **+2.33¢ per contract**

A Kalshi price is a probability, which removes a step: a 23¢ ask means the contract must hit more than 23% of the time to profit. No vig conversion, no odds arithmetic — compare the number to the price.

Where the per-point shortcut stops working

$1/(4W)$ is the slope of the curve *at the median*. The curve flattens away from there, so the linear rule overstates every step it takes into the tail. Within three points of the number the two agree to within 0.03 of a percentage point. By twelve points out the gap is four points of probability, and from the 6.5 rung inward the rule returns numbers above 100%, which is simply nonsense:

RUNG	EDGE (PTS)	PER-POINT RULE	EXACT	KALSHI ASK	AS ODDS	GAP (¢)
41.5	-15.86	6.5%	14.95%	17¢	+488	-2.05
38.5	-12.86	14.7%	19.62%	18¢	+456	+1.62
35.5	-9.86	23.0%	25.33%	23¢	+335	+2.33
33.5	-7.86	28.5%	29.69%	29¢	+245	+0.69
30.5	-4.86	36.7%	36.98%	36¢	+178	+0.98
27.5	-1.86	44.9%	44.91%	46¢	+117	-1.09
24.5	+1.14	53.1%	53.11%	53¢	-113	+0.11

RUNG	EDGE (PTS)	PER-POINT RULE	EXACT	KALSHI ASK	AS ODDS	GAP (¢)
23.5	+2.14	55.9%	55.83%	56¢	-127	-0.17
21.5	+4.14	61.3%	61.15%	60¢	-150	+1.15
20.5	+5.14	64.1%	63.72%	64¢	-178	-0.28
17.5	+8.14	72.3%	70.93%	71¢	-245	-0.07
16.5	+9.14	75.0%	73.14%	73¢	-270	+0.14
14.5	+11.14	80.5%	77.22%	77¢	-335	+0.22
13.5	+12.14	83.3%	79.09%	79¢	-376	+0.09
11.5	+14.14	88.7%	82.49%	82¢	-456	+0.49
10.5	+15.14	91.5%	84.01%	84¢	-525	+0.01
9.5	+16.14	94.2%	85.43%	86¢	-614	-0.57
7.5	+18.14	99.7%	87.96%	89¢	-809	-1.04
6.5	+19.14	102.4%	89.07%	90¢	-900	-0.93
5.5	+20.14	105.2%	90.09%	90¢	-900	+0.09
4.5	+21.14	107.9%	91.03%	90¢	-900	+1.03
3.5	+22.14	110.7%	91.88%	92¢	-1150	-0.12
2.5	+23.14	113.4%	92.66%	93¢	-1329	-0.34
1.5	+24.14	116.1%	93.38%	93¢	-1329	+0.38

Shaded rows are where the shortcut has drifted more than three points of probability from the truth. Use the exact logistic for anything past the rungs immediately around the number; keep the per-point rule for the main line, where it is exact enough and far easier to hold in your head.

The last column is exact – ask. Positive means the contract is worth more than it costs, so you would buy the YES: **14 of the 24 rungs** show that.

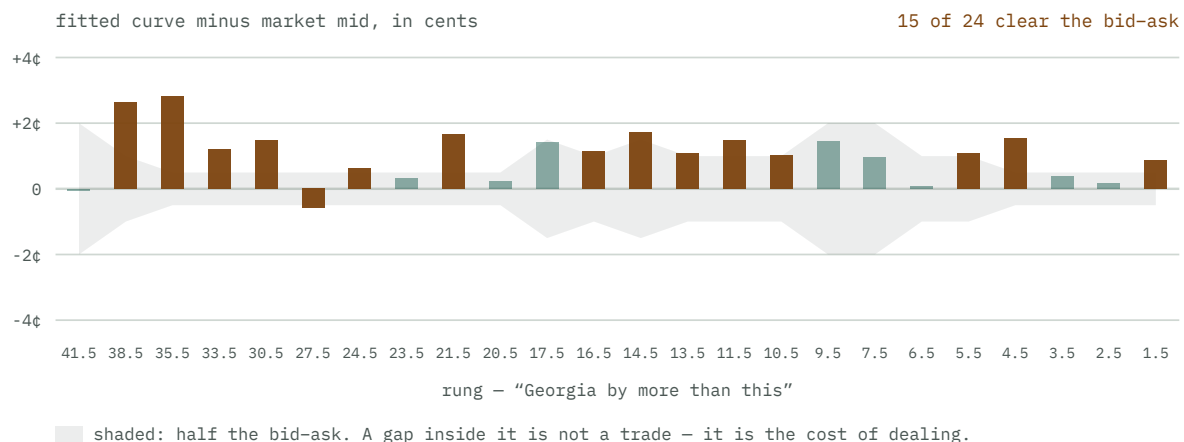
A negative number is *not* the mirror of it, and there are ten of them. All it says is that the model sits below the ask — and on nine of those ten it is still *above the bid*, which puts it inside the spread, where neither side is a trade. Selling needs the model below the **bid**, not merely below the ask, and exactly one rung manages that:

WHERE THE MODEL SITS	RUNGS	WHAT IT MEANS
above the ask	14	buy YES
below the bid	1	buy NO – the 27.5 rung, 44.91% against a 45¢ bid
between bid and ask	9	nothing to do

Which is why the chart below re-centres on the mid rather than the ask: it is the one view where both kinds of opportunity, and the nine non-opportunities, read off the same axis.

Visualized:

Both of those tests are one test, and it draws better re-centered on the mid price. A gap wider than half the bid-ask is exactly a row where the model beats the ask or undercuts the bid; a gap inside it is a row where the model sits between the two and there is nothing to do:



Each rung’s gap between the fitted curve and the market mid, against a shaded band of half the bid-ask. Bars inside the band are not opportunities — they are narrower than the cost of trading.

Fifteen of the 24 rungs show a gap wider than half the bid-ask, and the largest is 2.83¢, at the 35.5 rung. Nothing about that is a windfall: the board’s scale_residual for this game is 2.67¢, the same misfit measured from the ladder’s own median instead of the Vegas-shrunk estimate. The apparent opportunities and the misfit are one measurement, not two.

And the main rung makes the point sharply. Kalshi asks 53¢ for “Georgia by more than 24.5” — the identical bet the tool likes at Vegas –110:

our number 53.11%
 Vegas -110 needs 52.38% — clears by **0.73 pts**
 Kalshi 53¢ needs 53.00% — clears by **0.11 pts** = -113

Same game, same side, same estimate. Three cents of extra price eats 85% of the edge, and a one-cent-wider quote would erase it outright. What you pay is not a detail on top of the analysis; on a lean it is most of the analysis.

The cost of dealing is not flat

On Kalshi nobody pays -110; they pay the bid-ask. Buy the YES at the ask and the NO at the bid on a one-cent rung and the two sides sum to 101¢: a **1% overround**, against 104.76% for a pair of -110 sportsbook prices.

RUNG	BID	ASK	SPREAD	OVERROUND	VS -110
41.5	0.13	0.17	4¢	4.0%	0.84×
38.5	0.16	0.18	2¢	2.0%	0.42×
35.5	0.22	0.23	1¢	1.0%	0.21×
27.5	0.45	0.46	1¢	1.0%	0.21×
24.5	0.52	0.53	1¢	1.0%	0.21×
21.5	0.59	0.60	1¢	1.0%	0.21×
7.5	0.85	0.89	4¢	4.0%	0.84×
2.5	0.92	0.93	1¢	1.0%	0.21×

Most of this ladder costs about a fifth of what a Vegas spread costs — but the rungs that showed the biggest gaps are the wide ones. **The cheap rungs are the busy ones near the number; the far ones wanted for a tail bet are the expensive ones**, market makers charging for risk nobody else wants.

And the naive version of this trade is circular, fatally so. A Kalshi rung cannot be bet on the strength of a curve fitted to Kalshi rungs. The model is the market, smoothed, so every gap in that chart is the smoothing disagreeing with the prices it was built from. The logistic is a 3.7¢ approximation on a typical ladder, nearly six times the noise floor — so a 2.67¢ gap is far likelier to be the model being wrong than the market being wrong.

Where the comparison is *not* circular is what the tool actually does: bet a Vegas number using Kalshi as the signal. Two venues, two independently formed prices, and the disagreement means something.

Which points at a real extension. Sportsbooks sell **alternate lines** — Georgia -21.5 at -140, Georgia -31.5 at +150. Those are Vegas prices out at the far rungs, so comparing them against the Kalshi-fitted curve is not circular, and it is exactly the region where the fit has most to say. At present the tool compares only at the main number, spending a twenty-four-rung ladder to judge one number a week.

HOW FAR TO TRUST IT

The fit residual

Two parameters standing in for a whole market is a claim, and every game now ships the evidence: `scale_residual`, the worst gap between the fitted curve and the strikes it came from. On this ladder it is **0.0267** — nowhere in the 5–95¢ window does the fitted curve sit more than 2.7 cents from the market.

A residual only means something against a floor, so: generate prices from an *exact* logistic, degrade them the way Kalshi degrades everything — round to 1¢ ticks, add a realistic spread — and push them through the same pipeline.

```
prices from a known logistic, ticked and spread
residual = 0.006 ← the machinery measuring itself

real Kalshi ladders
residual = 0.037 median (5.8× the floor)
```

College football margins are not actually logistic. The deviation is nearly six times what tick rounding and spreads can explain, and in half the games the worst miss sits at the median rather than out in the tails. That is real shape, not noise.

Worth stating plainly, because it bounds the claim. Near the number 3¢ of probability is roughly a point of spread, so cover chance and the moneyline conversion carry about that much model error. It is an approximation, and a bounded one: every card prints its own residual rather than assuming the question away.

That number replaced a 140-value survival table each game used to ship purely so the browser could look the curve up. Cover chance is now closed form from a centre and a scale, nothing read the table, and dropping it took the payload from 131 KB to 101 KB.

Every figure is one live snapshot of Georgia at Arkansas, 16 September 2026, checked against the running pipeline — the survival points, the scale fit, the composite, the shrinkage and the cover probability all reproduce its published output exactly. Rebuild them from the ladder table and you will match to the last decimal, with the single documented exception of the median in step 3.